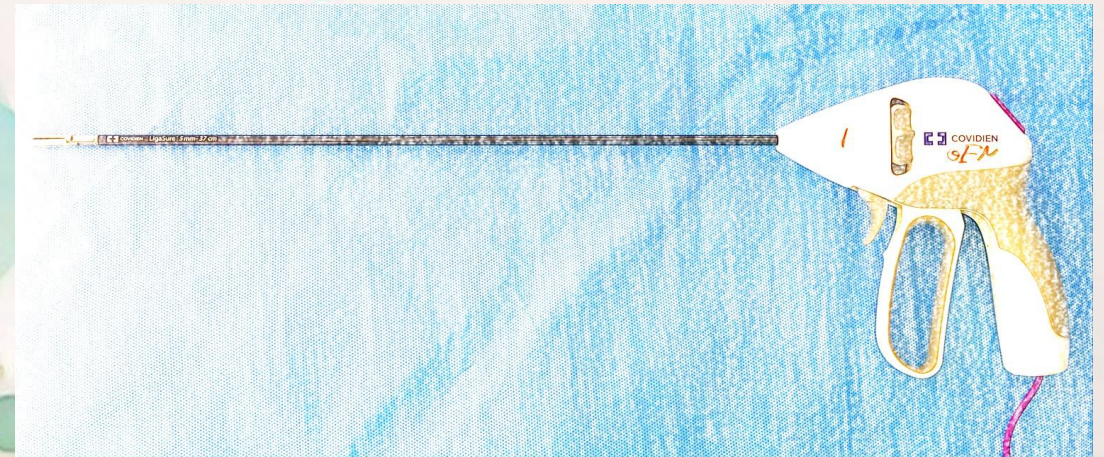


Perioperative Nursing care in GI surgery: Vascular sealing

Potchane Silchai R.N.



Overview of hemostasis

01

**Dissection, mechanical
compression, sutures, clips, staplers**



02

Electrosurgical tools

2.1 monopolar and bipolar electrocautery

2.2 vessel sealing devices



03

Tissue sealants



02

Electrosurgical tools

2.1 monopolar and bipolar electrocautery

2.2 vessel sealing devices



principle of high frequency surgery



High frequency current does not interfere with the body's normal function.

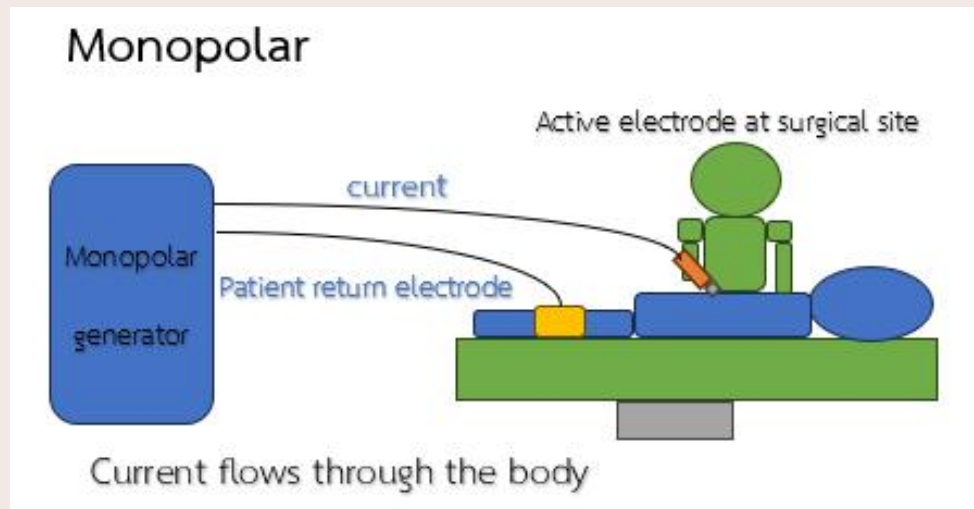


Low frequency current can cause electrocaution or cardiac arrest.



In HF surgery, high frequency alternating current (500 to 1000 kHz; 100 to 20000v) is used on patients

monopolar



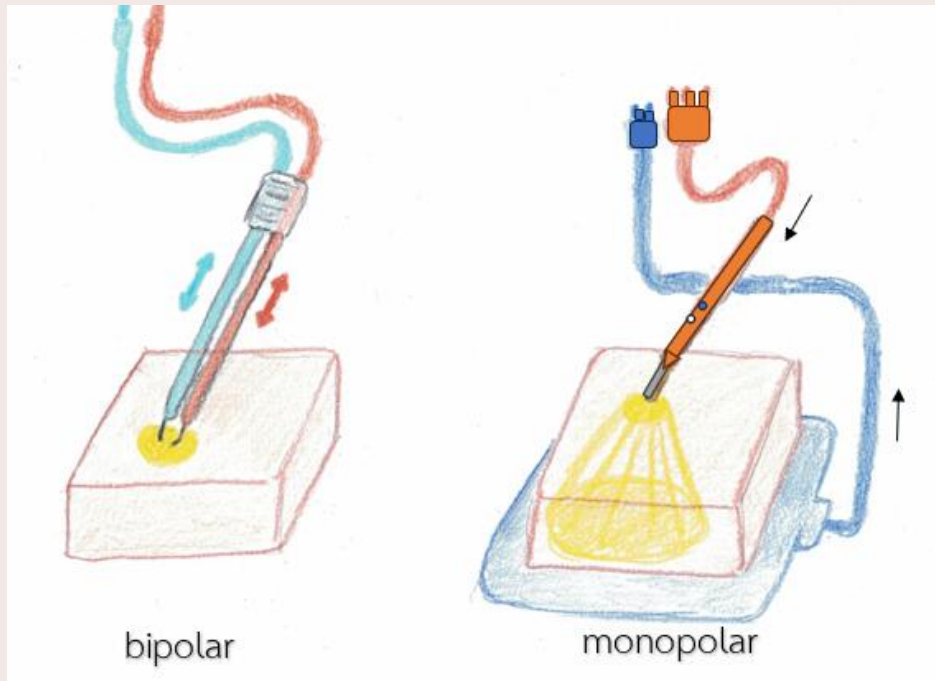
01 Easy to use

02 Multiple modalities

03 Inexpensive

04 Large thermal spread

bipolar

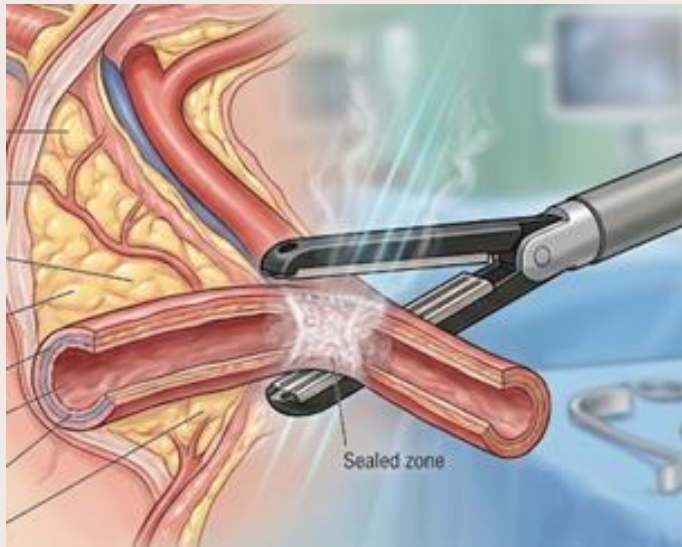


No current passes through the patient's body

No PRE is needed to disperse the current

Minimal heat is spread to surrounding tissues

Vessel sealing devices



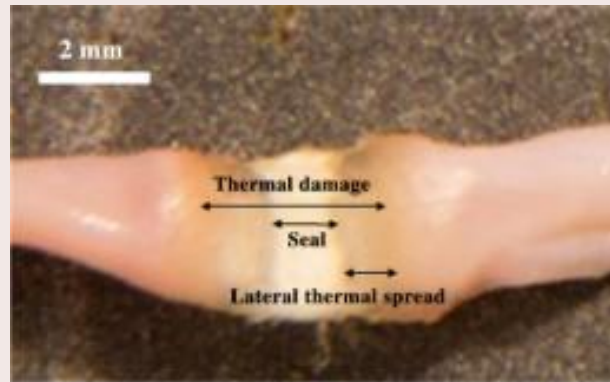
- **Bipolar technology that fuses collagen and elastin in the vessel walls and permanently obliterates the lumen of the vessel (AORN,2013)**
- **Can coagulate vessels and hemorrhagic tissues effectively and quickly**



Vessel sealing devices

Advantage

- To seal larger vessels (i.e, 5-7 mm diameter)
- Sutureless technique and operative time saving
- no use of foreign body material



ที่มา:
<https://www.spiedigitallibrary.org/>

Disadvantage

- High disposal cost
- Risk of lateral thermal spread injury

Vessel sealing devices

2 major categories

(Stephen D. Lyons, 2013)



**Advanced bipolar
devices**

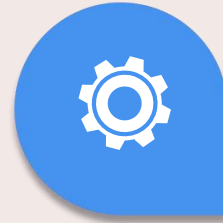


Ultrasonic devices

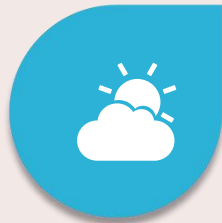
Advanced bipolar devices



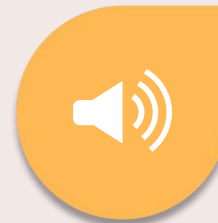
**A microprocessor
(programmable computer chip)**



**An alarm system that automatically
stops the current when the tissue
seal is achieved**



Tissue impedance monitoring



**Combines optimal thermal and
mechanical (graspers design)**



Advanced bipolar devices

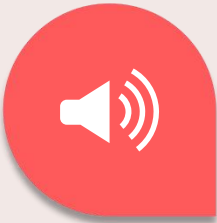
1.LigaSure™ Vessel Sealing System(Medtronic)

2.Enseal (Ethicon Endo-surgery)

3.PlasmaKinetic System (PKS;Gyrus ACMI)



1.LigaSure



First commercially available vessel sealing system (1998)



ForceTriad generator

- **4000 measurements of tissue impedance/second (conventional 200)**
- **Provide real-time adjustment control of energy output**
- **Shorter sealing time**



1.LigaSure



Does not work properly on tissue with staples

Disposable

model of this range with a cutting blade incorporated into the tip



2.EnSeal

The EnSeal G2 advanced bipolar device. This is the curved-tip model of this range (c.f., the straight-tip instrument). Note the retracted copper-colored transection blade at the jaw hinge.



ที่มา: <https://www.jnjmedtech.com/>



- I-Blade™ technology
- Seal and transect tissue, including vessels up to 7 mm in diameter



Minimal thermal spread ~ 1 mm
Minimal sticking



Produce little smoke & Lowest temperature (~100oC)



The ENSEAL™ X1 Curved Jaw Tissue Sealer



- 360-degree rotation
- separate seal and cut buttons

3.PKS



Pulsatile delivery of energy allows tissues to cool between energy bursts



2-6 mm. thermal spread



nonblades model of this range of advanced bipolar devices

BiClamp® (Erbe Elektromedizin GmbH)



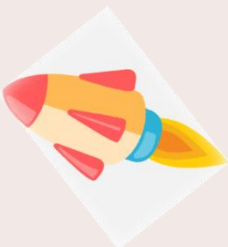
Reusable



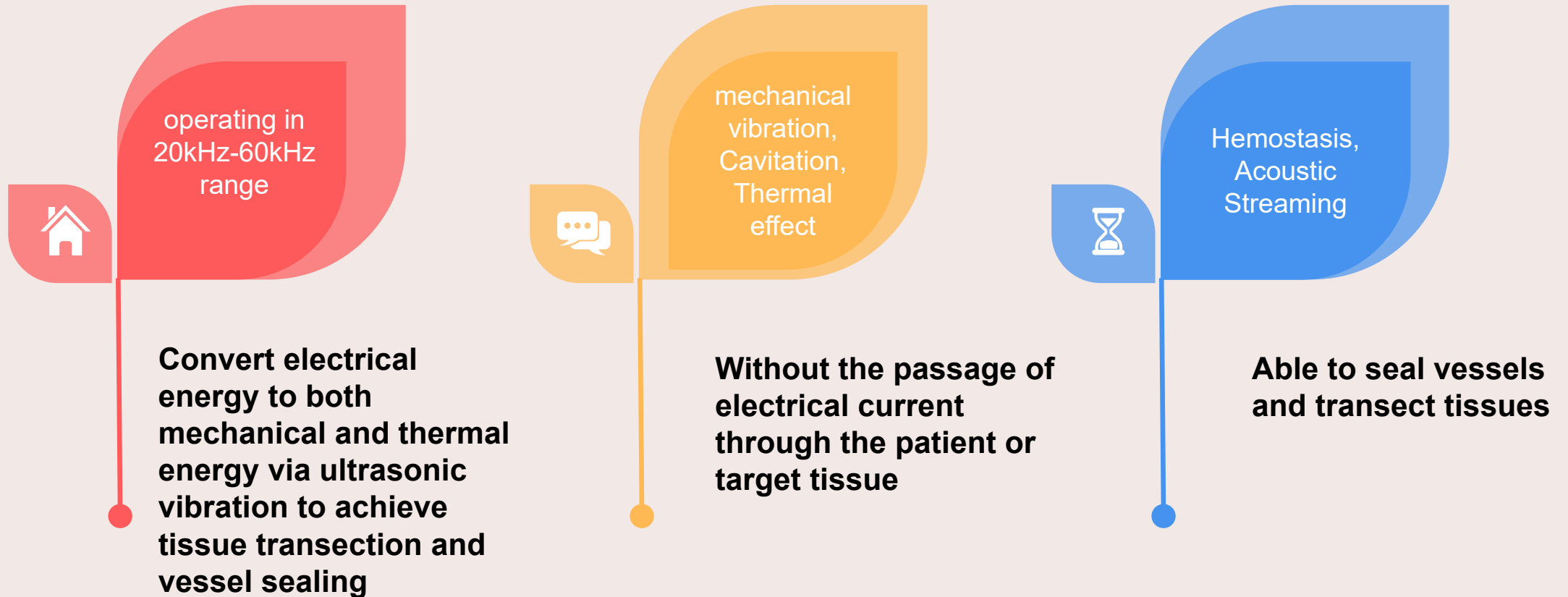
**ERBE VIO generator
automatically detects sealing end points**



**Ligates 2-7 mm^r arteries and
veins**



Ultrasonic Devices



Ultrasonic Devices

Advantages

01

Decrease
“instrument traffic”

02

Less tissue
necrosis and
charring

03

Less smoke
generation
compared with
electrosurgery

04

Reduce lateral
thermal spread



Ultrasonic Devices

Ultrasonic Hemostatic Dissectors

1. **Harmonic ACE**
(Ethicon Endo-Surgery)
2. **Sonocision** (Medtronic)

Ultrasonic Surgical Aspirators

The Cavitron Ultrasonic Surgical Aspirator
(CUSA)



Ultrasonic Hemostatic Dissectors

Harmonic scalpel

instrument



Handpiece wrench

generator



Harmonic scalpel

01

- **Cutting : sharp knife with frequency of 55,000 cycles/second**
- **Coagulation : vibration breaks hydrogen bonds & generates heat**

02

2 generator setting

- **Max rapid tissue transection: lateral thermal spread is less**
- **Min ideal for vessel sealing: increased risk of lateral thermal spread**



HARMONIC™ 1100 Shears



ที่มา: [https://www.https://www.jnjmedtech.com/](https://www.jnjmedtech.com/)

Ultrasonic Hemostatic Dissectors

Sonocision

cordless ultrasonic
device



Quick and simple set
up that increases O.R.
efficiency



Composed of single
use and reusable
component



Improve freedom of
movement



Sonocision

GENERATOR INDICATOR LIGHTS



[GREEN]

System is ready
(minimum or maximum power).



[RED]

Device is non-functional
(see intraoperative
troubleshooting section or IFU).



[FLASHING
YELLOW]

Battery charge is less than
20% capacity (approximately
10-20 activations remaining
upon initial notice).



[FLASHING
PURPLE]

Generator has reached end
of life (replace generator).

Ultrasonic Surgical Aspirators

The Cavitron Ultrasonic Surgical Aspirator (CUSA)



combines
ultrasonic energy
with aspiration



divide liver
parenchyma
(parenchymal
vessels & biliary >
2 mm



dissect tissue
(not hemostasis)



low blood loss



low risk of bile leak



Original Article

A retrospective comparative study of hemorrhoidal artery ligation versus ligasure hemorrhoidectomy for the third degree hemorrhoidal disease

Tolga Onder ^a , Merih Altıok ^b 

classical surgery

- intense pain
- bleeding
- long recovery time

Ligasure™ hemorrhoidectomy & DGHAL (Doppler-guided hemorrhoidal artery ligation)

- less pain
- less bleeding
- shorter hospital stays
- low complication rates

Review

Advances in Minimally Invasive Esophagectomy—An Overview of Recent Developments and a Novel Classification of Innovations in Treatment of Thoracic Esophageal Cancer

Florin Achim ^{1,2,*} , Koji Otsuka ^{3,4} , Takeshi Yamashita ^{3,5}, Yutaro Asagoe ⁶, Daisuke Kurita ⁷, Adrian Constantin ^{1,2,*}, Silviu Constantinoiu ^{1,2}, Ahmed Mohssen ⁸, Cristian Rosianu ⁸, Alexandru Rotariu ^{1,2}, Alex-Claudiu Moraru ^{1,2}, Anthony Rasuceanu ^{1,2} and Dragos Predescu ^{1,2}

Medicina **2025**, *61*, 2176

<https://doi.org/10.3390/medicina61122176>

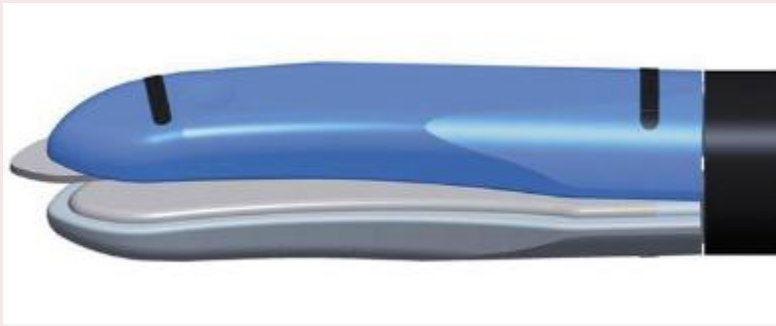


Bipolar vessel sealing VS ultrasonic devices in thermal damage of laryngeal nerve

Bipolar vessel sealers : superior protection against in advertent thermal damage to critical mediastinal structures in some reports (mediastinal dissection and lymphadenectomy along the RLN)

Ultimately, the choice of instrument depends on the surgeon's preference, experience, and availability

Combine vascular sealing



Ligasure Advance,
Medtronic



Monopolar and
bipolar technology



Smoke production
affecting visibility

Combine vascular sealing



Thunderbeat, Olympus

Ultrasonic and bipolar technology

Seal & Cut Mode

Bipolar energy and ultrasonic energy

Seal

Advanced bipolar energy

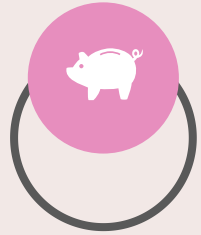
ที่มา: <https://medical.olympusamerica.com>

Advance bipolar & Ultrasonic Devices

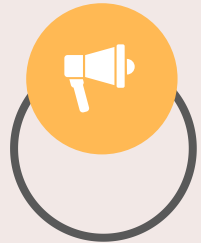
	ตารางเปรียบเทียบคุณลักษณะสำคัญ	
	Ligasure (Advance Bipolar)	Ultrasonic Devices
กลไกการทำงาน	ใช้พลังงานไฟฟ้า bipolar ร่วมกับแรงกดอัดเนื้อเยื่อ หลอมรวม collagen และ elastin ในผนังหลอดเลือด	ใช้การสั่นสะเทือนทางกลความถี่สูง (20-60 kHz) สร้างความร้อนจากการเสียดสี เกิด Cavitation และ denature ของโปรตีน
อุณหภูมิสูงสุด (Blade/Jaw)	ต่ำกว่า ใบมีด ~81-95.6 °C กรามหนีบ ~96-115 °C	สูงกว่ามาก ใบมีด ~115-330 °C กรามหนีบ ~168-343 °C
ระยะเวลาระบายความร้อน (ให้เหลือ <60 °C)	เร็วกว่า 2-3 เท่า ~13.4-27.4 วินาที	ช้ากว่า ~28.7-95.7 วินาที
การแพร่กระจายความร้อนออกด้านข้าง (Lateral thermal spread)	แคบกว่า (~0.56-0.75 มม.)	กว้างกว่า (~1.29-2.39 มม.)
ปริมาณควันผ่าตัด (Surgical Plume)	มีควันเล็กน้อยจากการจัดความร้อน	เกิดควันน้อยมาก ช่วยให้ทัศนวิสัยชัดเจนในช่องท้อง/ช่องอก

Perioperative nursing care

Pre-operative phase



Know which ESU will be used and how to use it.



Have all equipment and accessories available



The ESU should be tested before initial use

Perioperative nursing care

Intraoperative phase



Inspect instruments

The instruments should be visually inspected at the surgical field, before use,

- any apparent damage (eg, impaired insulation)

Test instruments

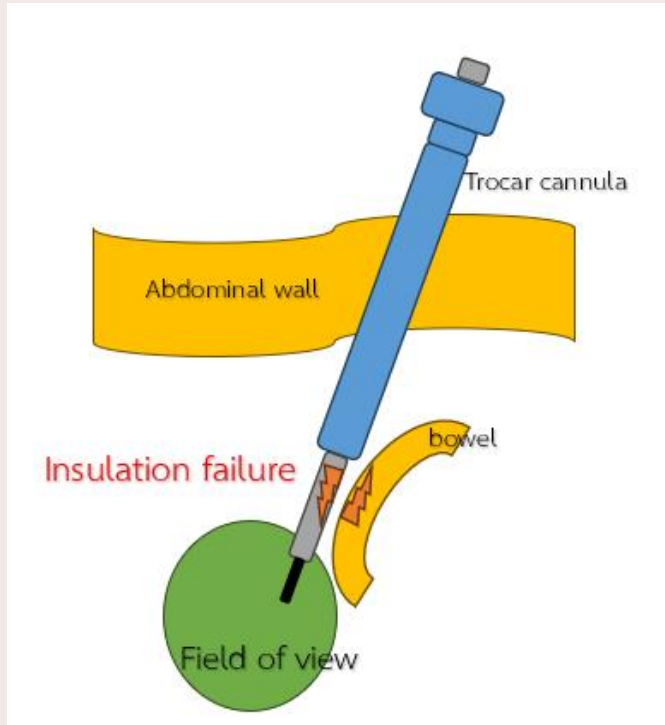
Connect the instrument power cord to the generator, test instruments

Volume level

Do not turn the activation tone down to an inaudible level.

Power setting

Use the lowest possible power settings



Perioperative nursing care

Intraoperative phase

when not in use



place electrosurgical
instruments in a safety
holster



away from
patients, the
surgical team
and flammable
materials



Prevent accidental
burns (Ultrasonic
scalpel blades
remain hot
immediately after
use)



Unintentional
activation
cause patient
and personnel
injury

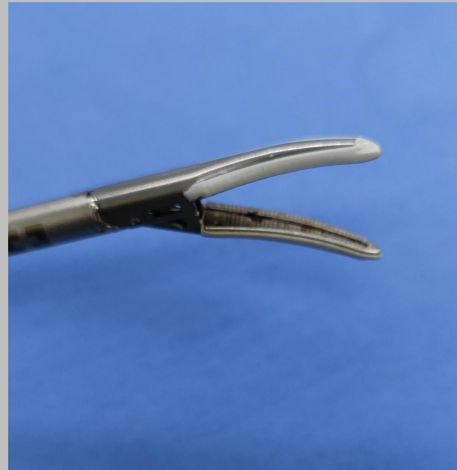


Perioperative nursing care

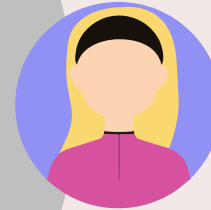
Intraoperative phase



The instrument can be wiped with a sterile moist gauze sponge to remove tissue



Do not activate the instrument while cleaning to avoid inadvertent user injury



Do not clean the instrument with abrasive

Perioperative nursing care

Intraoperative phase



Do not immerse distal tip in liquid including saline solution to clean (bipolar)



At the conclusion of the case dispose the instruments in appropriate container



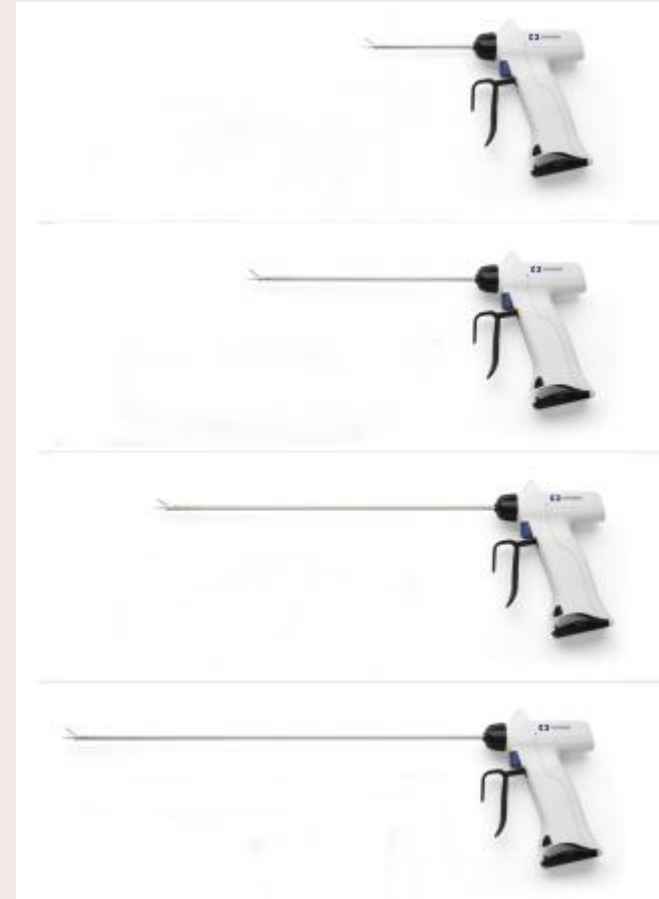
Consult the pacemaker manufacturer/cardiology department for information for use of electrosurgery or fusion appliances is planned in patient with cardiac pacemaker

conclusion

Knowing what type of energy?

Safely use

- Visually inspected
- test before use
- Prevention of injury to patients and personnel (accidental burn, clean)





THANK YOU

